Student Name: Elum Joy

Registration Number: C11/26/PCDFIR/17031

Course Code: CIP – B101

Lab Title: PC System Fundamentals, Disk Sectors and Partition Analysis

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Instructor Name: Abdullahi Umar Abbas

Introduction

The purpose of this lab is to understand how digital evidence is created and stored on a computer system. It begins by mapping evidence across system layers: applications, operating system, file system, disk, memory, and network. Next, it explains the boot process from power-on to operating system control and identifies whether my system boots using BIOS/Legacy or UEFI. It then identifies the file systems on my system's volumes and explains why they matter in digital forensics. Finally, it verifies partition sizes using partition table evidence, by converting sector counts to gigabytes and comparing the results with OS-reported sizes. Together, these tasks show how forensic examiners locate, interpret, and verify evidence at the system level.

Tools used:

1. Windows operating system (windows 10)
2. Windows PowerShell (Administrator)
3. System Information (msinfo32)
4. Windows Snipping Tool / screenshots
5. Disk Management (diskmgmt.msc)
6. diskpart

Commands Used:

1. Get-Disk -Number 0 | Select-Object LogicalSectorSize
2. diskmgmt.msc
3. diskpart, then select disk 0, then list partition
4. Get-Partition -DiskNumber 0 | Format-Table PartitionNumber,Type,Offset,Size -AutoSize
5. Get-Partition -DiskNumber 0 -PartitionNumber 2 | Select-Object PartitionNumber,GptType,Type
6. Get-Partition -DiskNumber 0 -PartitionNumber 1 | Select-Object PartitionNumber,GptType,Type
7. Get-Partition -DiskNumber 0 | Format-Table PartitionNumber,Type,Offset,Size -AutoSize

Software used:

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6. diskpart

A.1. Basic system specification

```
C:\Users\Joy Elum>systeminfo

Host Name:                 JUSTADA
OS Name:                   Microsoft Windows 11 Education
OS Version:                10.0.26300 N/A Build 26300
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          Joy Elum
Registered Organization:   N/A
Product ID:                00328-00003-42875-AA140
Original Install Date:     11/6/2024, 11:28:16 AM
System Boot Time:          10/31/2026, 6:33:01 PM
System Manufacturer:       Dell Inc.
System Model:              Latitude 7490
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 142 Stepping 10 GenuineIntel ~1508 Mhz
BIOS Version:              Dell Inc. 1.41.0, 11/6/2024
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-gb;English (United Kingdom)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+01:00) West Central Africa
Total Physical Memory:     16,262 MB
Available Physical Memory: 8,488 MB
Virtual Memory: Max Size:  23,430 MB
Virtual Memory: Available: 14,807 MB
```

A.2. System Storage Information

Item	Value
Hardware Abstraction Layer	Version = "10.0.26100.1"
Username	JUSTADA\Joy Elum
Time Zone	W. Central Africa Standard Time
Installed Physical Memory (RAM)	16.0 GB
Total Physical Memory	15.9 GB
Available Physical Memory	8.20 GB
Total Virtual Memory	22.9 GB
Available Virtual Memory	14.5 GB
Page File Space	7.00 GB
Page File	C:\pagefile.sys
Kernel DMA Protection	Off
Virtualisation-based security	Running
Virtualisation-based security re...	Base Virtualisation Support
Virtualisation-based security av...	Base Virtualisation Support, DMA Protection, UEFI Code Readonly, Mode Based...
Virtualisation-based security ser...	Hypervisor enforced Code Integrity
Virtualisation-based security ser...	Credential Guard, Hypervisor enforced Code Integrity
App Control for Business policy	Enforced
App Control for Business user m...	Off
Automatic Device Encryption Su...	Elevation Required to View
A hypervisor has been detected....	

A.3.

Map evidence sources to system layers

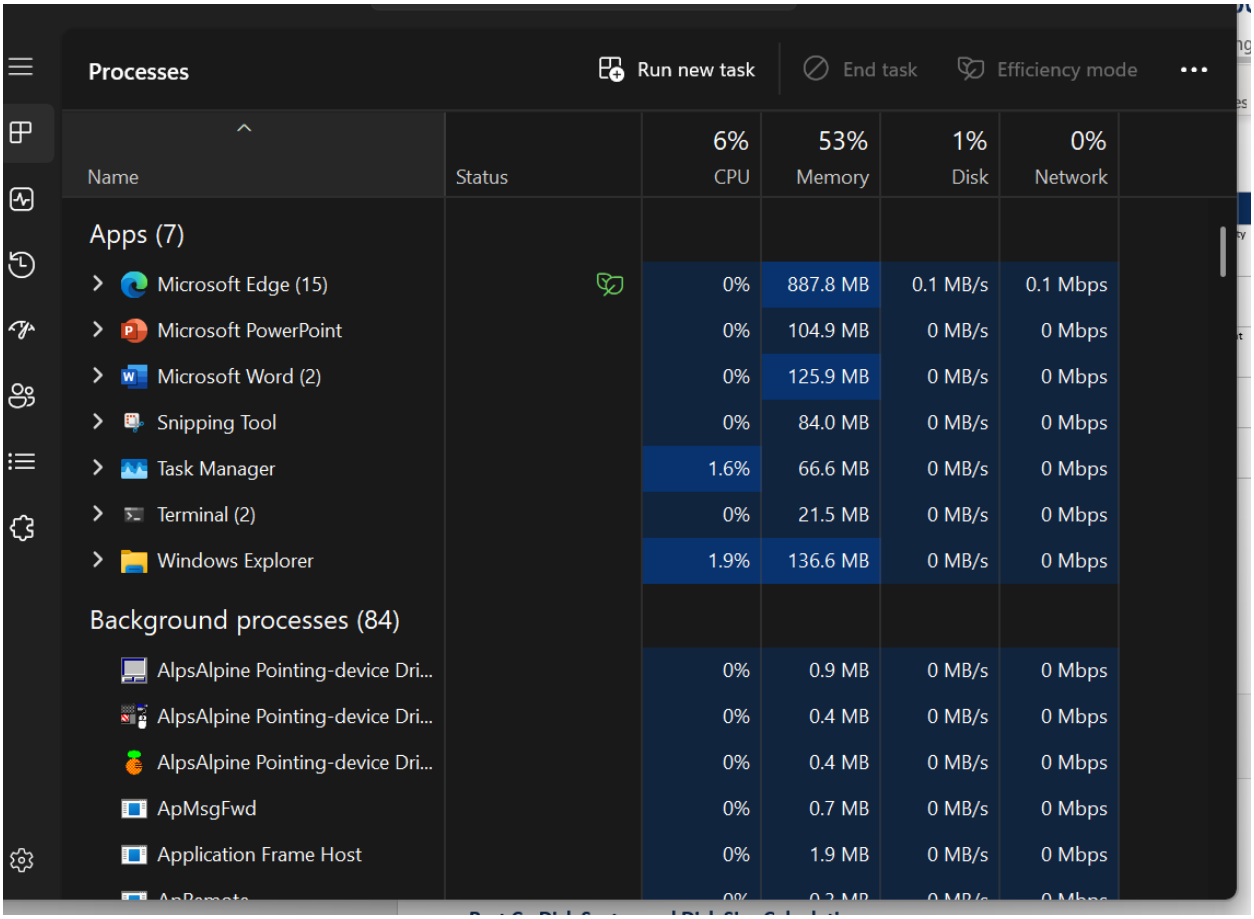
	Layer	Sources	Type of Evidence Generated
1	Applications	Web browsers, email clients, messaging apps (WhatsApp, Telegram), office software, cloud sync clients	Browsing history, cache, cookies, downloads, chat logs, sent/received emails, document metadata, recent-file lists
2	Operating System	Windows Registry, event logs, user profiles, prefetch files, shellbags, jump lists, system logs (e.g., /var/log)	Login/logout times, program execution history, USB device connections, user account activity, file access traces, system configuration changes
3	File System	NTFS ($MFT, $UsnJrnl), ext4 journal, file allocation tables, Recycle Bin	File creation, modification, access and deletion timestamps, file names and paths, deleted file records, folder structure

4	Disk (Storage)	Hard drive, SSD, USB drives, unallocated space, slack space, hibernation file	Deleted or hidden files, file fragments, partition data, encrypted volumes, remnants of overwritten data
5	Memory (RAM)	Live RAM, page/swap file, crash dumps	Running processes, open network connections, encryption keys, passwords in clear text, injected malware, unsaved documents, clipboard contents
6	Network	Router and firewall logs, DNS logs, proxy servers, ISP records, packet captures	Connection history, IP addresses, domains visited, data transfer volumes, remote access and lateral movement traces

Reasons evidence may exist in more than one layer:

1. **One action touches many layers.** When a user opens a file, the application records it in a recent-files list, the operating system logs execution and access, the file system updates timestamps, and the disk or memory holds the content itself.
2. **Data is copied and cached by design.** Systems write temporary copies to cache, swap files, and journals for speed and recovery, which leaves duplicate traces.
3. **Deletion is rarely complete.** Deleting a file removes the pointer at the file system layer, but the data often remains on disk, and references remain in application history and OS artifacts.
4. **Volatile and persistent data overlap.** Data in RAM can be written to disk through paging, hibernation, or crash dumps, so it may persist after the system shuts down.
5. **Corroboration strengthens reliability.** Matching evidence across layers confirms timelines and shows that an artifact was not tampered with. If an attacker alters one layer (for example, clearing logs), traces in another layer can still expose the activity.

B.1. Running Process evidence



B.2. Running Processes on the machine being used, their PID and likely purpose

S/N	Process	PID	Likely Purpose
1	smss.exe	1420	First user-mode process Windows starts at boot. It creates sessions and launches
2	Wininit.exe	700	Starts key system processes such as services.exe and lsass.exe. It runs in session 0 during startup.
3	fontdrvhost.exe	1852	Runs the font driver in user mode, away from the kernel, so a malformed font can't crash or compromise the system.

B.2i. Difference Between Program, process and thread

Program: A program is a file stored on disk that holds instructions but isn't running. notepad.exe sitting in a folder is a program. It uses no memory or CPU.

Process: A process is a program that has been loaded into memory and is running. The operating system gives it its own PID, memory space and resources. If you open Notepad twice, you get two processes from one program.

Thread: A thread is the smallest unit of work inside a process. A process has at least one thread, and many have several. In a browser, one thread might load a page while another plays audio. Threads in the same process share its memory, so they work together quickly.

B.2ii. **Why process information matters in malware analysis and live forensic triage**

1. **It shows what is running right now.** Process lists, PIDs and network connections exist only while the machine is on. They disappear at shutdown, so they are volatile evidence.
2. **It helps spot malware.** Suspicious signs include:
    - a fake name close to a real one, such as svch0st.exe
    - a system process running from the wrong folder
    - an unexpected parent process
    - unusually high CPU or network use
3. **It shows who is running what.** The user account start time and linked connections tie activity to a person and a timeline.
4. **It guides what to capture.** Triage tells the investigator which processes to dump from memory before the system is shut down.

Part C - Disk Sector and Disk Size Calculation

i.Disk Details

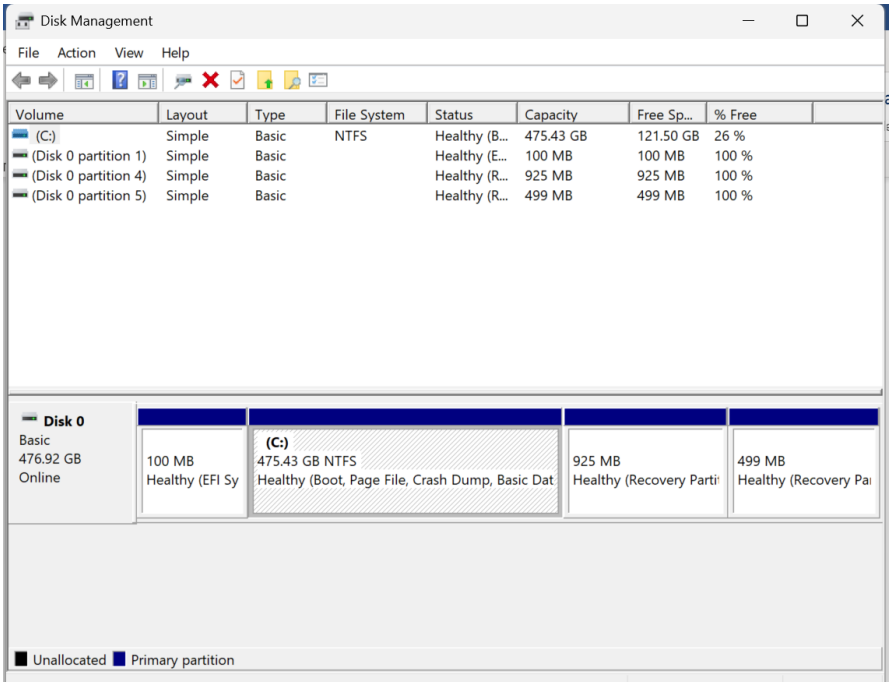
Partition	Type	Start sector	Number of sectors	Sector size
0	GPT: System (EFI)	1,048,576 ÷ 512 = 2,048	204,800	512
1	GPT: Basic Data (C:)	122,683,392 ÷ 512 = 239,616	997,054,464	512
2	GPT: Unknown (Recovery)	510,614,568,960 ÷ 512 = 997,294,080	1,894,400	512
3	GPT: Unknown (Recovery)	511,585,550,336 ÷ 512 = 999,190,528	1,021,952	512

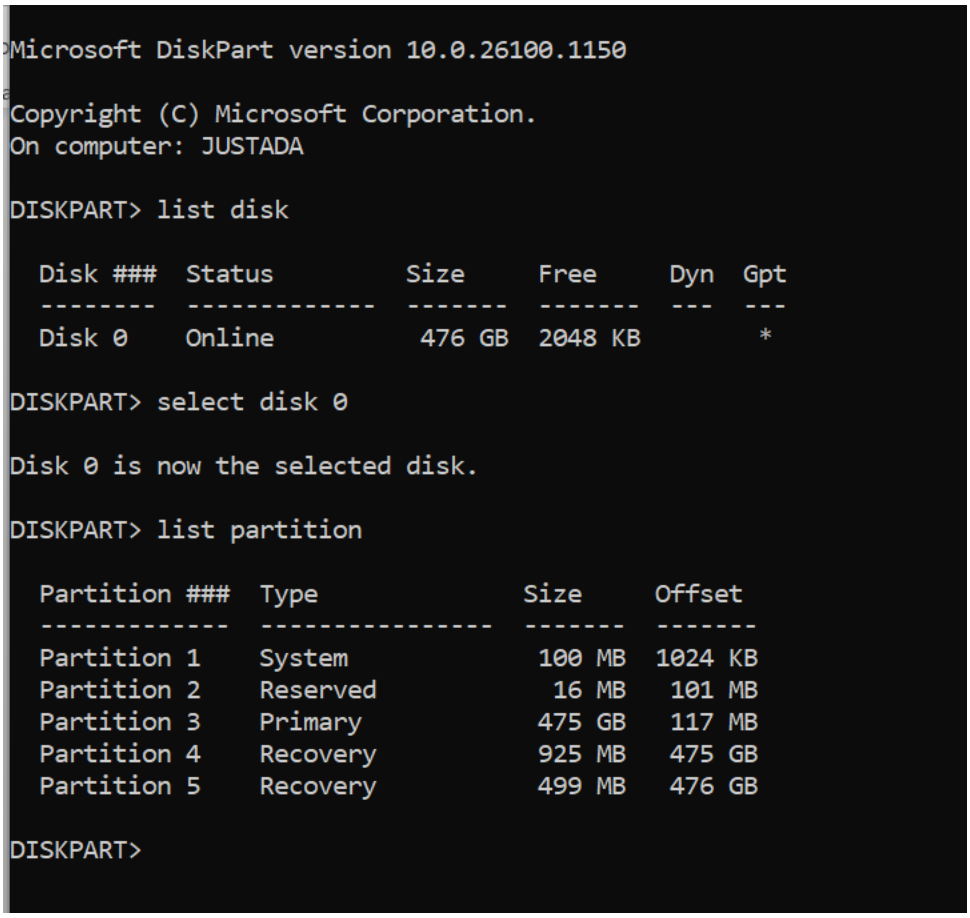
ii. Size of each partition in GB. Formula=sectors × 512 ÷ 1024 ÷ 1024 ÷ 1024

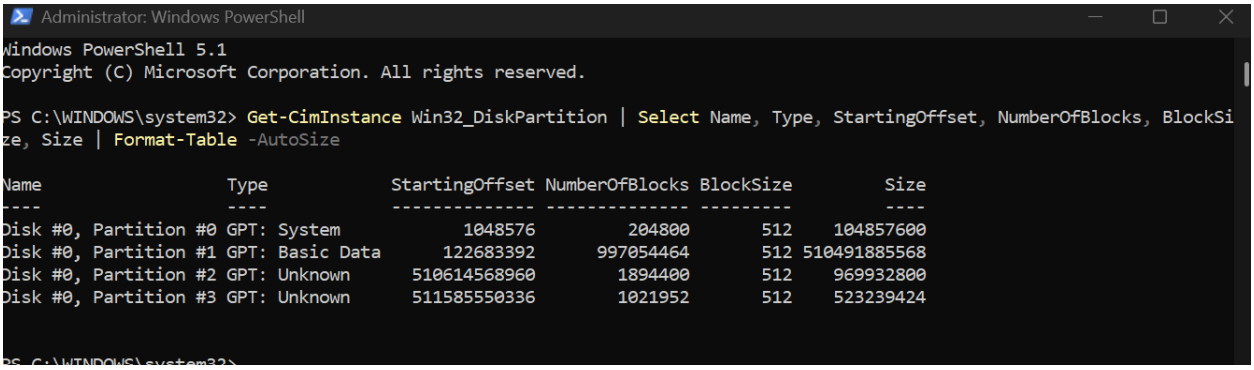
Partition	Calculation	Result	Disk Management
0	204,800 × 512 = 104,857,600 bytes; ÷ 1,073,741,824	0.0977 GB (100 MB)	100 MB
1	997,054,464 × 512 = 510,491,885,568 bytes; ÷ 1,073,741,824	475.43 GB	475.43 GB
2	1,894,400 × 512 = 969,932,800 bytes; ÷ 1,073,741,824	0.9033 GB (925 MB)	925 MB
3	1,021,952 × 512 = 523,239,424 bytes; ÷ 1,073,741,824	0.4873 GB (499 MB)	499 MB

Part D - Safe Partition Table Inspection

D.1. Windows method - Disk Management, PowerShell and Disk Partion







#D.3. This was optional part and was not performed to avoid any risk to the live system disk. GPT was confirmed through PowerShell (Get-Disk), which reports Partition Style GPT, and all partitions were listed in the supporting screenshots above.

Part E - MBR, GPT and Boot Process Explanation

E.1. Explain the boot process

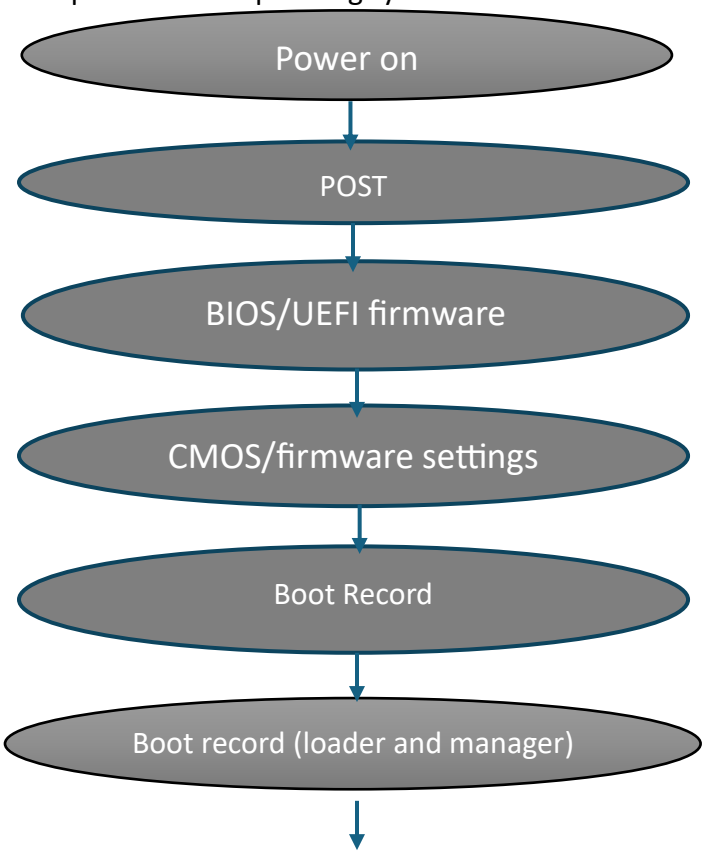
When the PC is powered on, the power supply stabilizes and the CPU resets. It starts running firmware code stored on the motherboard chip. The firmware runs the Power-On Self-Test which checks the CPU, memory, and storage controllers. If a component fails, the system shows beep codes or an error.

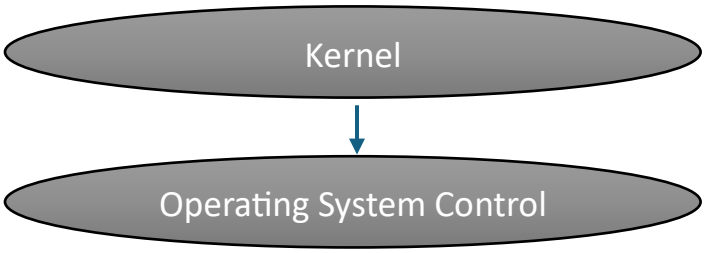
After POST, the BIOS or UEFI firmware (in my case BIOS) initializes the hardware and reads its stored CMOS/firmware settings, such as boot order, date and time, and Secure Boot. These settings decide which device the system boots from. They are kept in non-volatile memory powered by the CMOS battery.

The firmware then looks for the on the chosen disk. A BIOS system reads the MBR, the first 512-byte sector, which holds boot code and the partition table. A UEFI system reads the GPT and the EFI System Partition, then runs the boot file listed in its settings.

That file starts the boot loader or boot manager, such as Windows Boot Manager or GRUB. The boot loader locates the operating system and loads the kernel into memory, along with the drivers needed to start. The kernel then initializes memory management, drivers, and system services. At that point the operating system takes control of the machine and shows the login screen.

Fig; PC boot process from power-on to operating system control.





Each stage leaves evidence, such as boot records, ESP contents, and firmware settings, and tampering (bootkits) happens at stages 5 to 7.

#Personal system information evidence screenshot

Processor	Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz, 2112 Mhz, 4 Core(s), 8 Logical Pro...
BIOS Version/Date	Dell Inc. 1.41.0, 11/6/2024
SMBIOS Version	3.1
Embedded Controller Version	255.255
BIOS Mode	UEFI
BaseBoard Manufacturer	Dell Inc.
BaseBoard Product	02766V
BaseBoard Version	A10
Platform Role	Mobile
Secure Boot State	Off
PCR7 Configuration	Elevation Required to View

System Information (msinfo32) shows BIOS Mode is UEFI confirming my system boots using GPT with an EFI System Partition and Secure Boot is off as at the time

Part F - File System Identification and Interpretation

Evidence:

```
PS C:\Users\Joy Elum> Get-volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         NTFS           Fixed     Healthy      OK                    485.04 MB    499 MB
                         NTFS           Fixed     Healthy      OK                       108 MB    925 MB
C                        NTFS           Fixed     Healthy      OK                    121.03 GB 475.43 GB
```

The PowerShell command (Get-Volume) shows three fixed volumes, all formatted as NTFS, including the primary OS volume C: (475.43 GB). The two smaller volumes have no drive letter and appear to be system/recovery partitions.

The file system used by two volumes or partitions on your system.

Volume	File System	Size	Likely Role
C:	NTFS	475.43 GB (121.03 GB free)	Windows OS and user data
Unlabelled(no drive)	NTFS	925 MB (108MB free)	Hidden system or recovery partition

It was established that volumes on the system use NTFS file system

Why a file system is important in digital forensics:

1. It determines how data is organized. The file system maps file names, folders, and locations on disk, so an investigator must know it to find and read data.
2. It stores metadata which could be Timestamps created, modified or accessed, file sizes, owners, and permissions come from the file system. NTFS keeps these in the $MFT, which helps build timelines.
3. It affects deleted-file recovery as deleting a file usually removes only its record or pointer, so the data often remains in unallocated space but how recoverable it is, is dependent on the file system.
4. It determines which artifacts and tools apply
5. Misidentifying the file system leads to wrong timestamps, missed hidden data, or failed evidence extraction, which weakens the case.

Part G - Final Practical Assignment

1. **Identify Partition one**

Required evidence

```
PS C:\WINDOWS\system32> Get-Partition -DiskNumber 0 | Format-Table PartitionNumber,Type,Offset,Size -AutoSize

PartitionNumber Type           Offset         Size
--------------- ----           ------         ----
              1 System         1048576    104857600
              2 Reserved     105906176     16777216
              3 Basic        122683392 510491885568
              4 Recovery 510614568960    969932800
              5 Recovery 511585550336    523239424
```

The Get-Partition -DiskNumber 0 output shows partition 1 as Type **System** (the EFI System Partition) on Disk 0, which uses a GPT partition table. It starts at byte offset **1,048,576** and has a size of **104,857,600 bytes**

2. **Record partition 1 sector count**

Required evidence

```
PS C:\WINDOWS\system32> Get-Disk -Number 0 | Select-Object LogicalSectorSize

LogicalSectorSize
-----------------
              512
```

Partition 1 (System/EFI) has a byte offset of 1,048,576 and a size of 104,857,600 bytes. With a logical sector size of 512 bytes, the start sector is 1,048,576 ÷ 512 = **2,048**. The sector count is 104,857,600 ÷ 512 = **204,800 sectors**. The end sector is 2,048 + 204,800 − 1 = **206,847**

3. Calculate partition 1 size

Evidence Required

Step	Formula	Result
Size in bytes	204,800 sectors × 512	104,857,600 bytes
Size in GB (decimal)	104,857,600 ÷ 1,000,000,000	0.105 GB

4. Identify Partition 2

Evidence Required

```
PS C:\WINDOWS\system32> Get-Partition -DiskNumber 0 | Format-Table PartitionNumber,Type,Offset,Size -AutoSize

PartitionNumber Type           Offset         Size
--------------- ----           ------         ----
              1 System         1048576    104857600
              2 Reserved     105906176     16777216
              3 Basic        122683392 510491885568
              4 Recovery 510614568960    969932800
              5 Recovery 511585550336    523239424
```

```
PS C:\WINDOWS\system32> Get-Partition -DiskNumber 0 -PartitionNumber 2 | Select-Object PartitionNumber,GptType,Type

PartitionNumber GptType                              Type
--------------- -------                              ----
              2 {e3c9e316-0b5c-4db8-817d-f92df00215ae} Reserved
```

Command prompt;

Get-Partition -DiskNumber 0 -PartitionNumber 2 | Select-Object PartitionNumber,GptType,Type

5. **Record Partition 2 sector count**

Item	Formula	Result
Start sector	105,906,176 ÷ 512	206,848
Sector count	16,777,216 ÷ 512	32,768
End sector	206,848 + 32,768 – 1	239,615

Partition 2 (Reserved) has a byte offset of 105,906,176 and a size of 16,777,216 bytes. With a logical sector size of 512 bytes, the start sector is 105,906,176 ÷ 512 = **206,848**. The sector count is 16,777,216 ÷ 512 = **32,768 sectors**. The end sector is 206,848 + 32,768 − 1 = **239,615**

6. **Calculate partition 2 size**

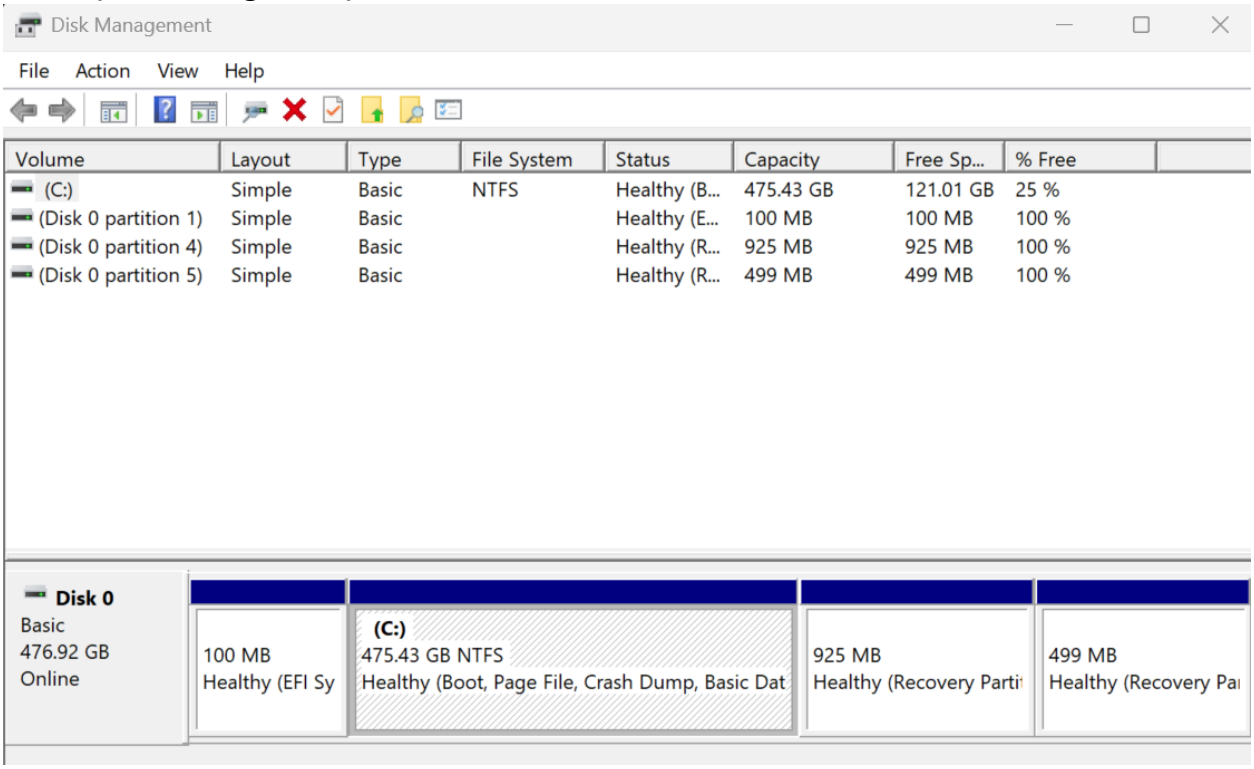
Step	**Formula**	**Result**
Size in bytes	32,768 sectors × 512	16,777,216 bytes
Size in GB (decimal)	16,777,216 ÷ 1,000,000,000	0.0168 GB

7. **Compare with OS-reported size**
   i. **Partition 1**

Calculation: 104,857,600 bytes = 0.105 GB

OS reported: 100mb

Source(Disk Management):



ii. Partition 2

Calculation: 16,777,216 bytes = 0.0168 GB

OS reported: 16mb

Source(Diskpart on powershell):

```
DISKPART> list partition

  Partition ###  Type              Size     Offset
  -------------  ----------------  -------  -------
  Partition 1    System             100 MB  1024 KB
  Partition 2    Reserved            16 MB   101 MB
  Partition 3    Primary            475 GB   117 MB
  Partition 4    Recovery           925 MB   475 GB
  Partition 5    Recovery           499 MB   476 GB

DISKPART>
```

Conclusion

In this lab, I learned that digital evidence is not stored in one place. It is spread across system layers though applications, operating system, file system, disk, memory, and network, and one user action can leave traces in several of them. By tracing the boot process from power-on to operating system control, I saw where firmware, boot records, and boot loaders fit, and I confirmed that my system boots using UEFI with a GPT disk. When I identified the file systems on my system, I found that three volumes, including the C: drive and two recovery partitions, use NTFS, which keeps timestamps, permissions, and journal records that investigators depend on.

In the partition table exercise, I used the sector size of 512 bytes to calculate that partition 1, the EFI System Partition, has 204,800 sectors, which is 104,857,600 bytes, and partition 2, the Microsoft Reserved Partition, has 32,768 sectors, which is 16,777,216 bytes. I then checked these against two Windows tools. Disk Management reported partition 1 as 100 MB, and diskpart reported partition 2 as 16 MB. Diskpart's offsets of 1024 KB and 101 MB also matched my partition table.

This matters in digital forensics because I cannot rely on a single source when I locate and interpret evidence. Partition tables and file systems decide where data lives and how it is read, so a mistake there can cause me to miss evidence or misreport it. Cross-checking my calculated results against more than one tool makes my findings more accurate and easier to defend.